

AR / MLM merge diagnostic in the V_d response space

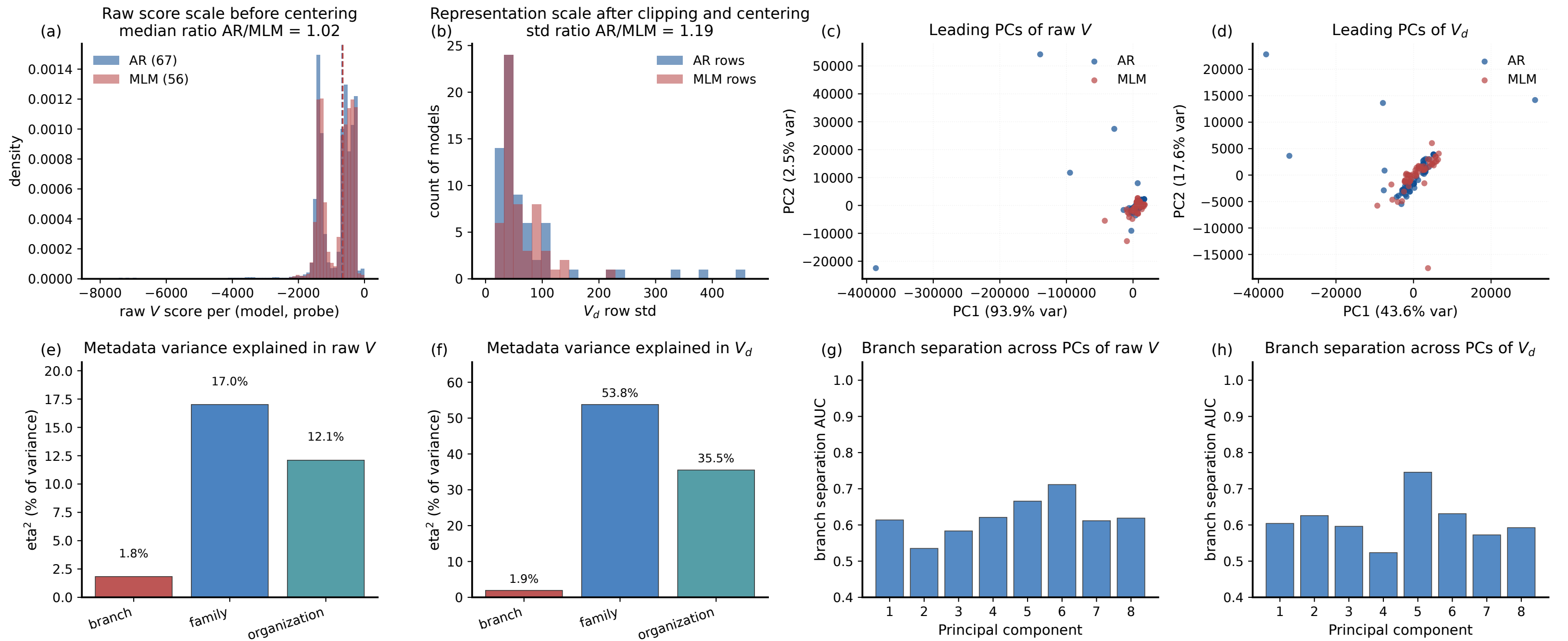


Figure. Empirical diagnostic for jointly analyzing autoregressive (AR) and masked-language-model (MLM) genomic language models in GLMap. For each model and probe, raw V stores the branch-native score: AR models contribute sequence log-likelihoods, whereas MLM models contribute pseudo-log-likelihoods. Because these raw scores are not the same probability object, GLMap compares their response profiles after lower-tail clipping and double-centering, yielding V_d . (a) Raw-score distributions show that AR and MLM marginal score scales are not separated by orders of magnitude (median ratio shown), but this panel is only a scale sanity check rather than evidence of probabilistic equivalence. (b) Row-standard-deviation distributions in V_d compare the residual response amplitude of each model after normalization; similar AR and MLM spreads indicate that neither branch dominates downstream distances simply by having a larger response scale. (c) PCA of raw V shows that the unnormalized matrix is dominated by a single global score axis, consistent with strong calibration or probe-difficulty effects in raw likelihood space. (d) PCA of V_d shows the centered GLMap response space after those first-order model-wide and probe-wide effects are removed. (e) Eta² decomposition in raw V quantifies how much total matrix variance is explained by branch, family, and organization labels. (f) The same eta² decomposition in V_d shows that the AR/MLM branch label explains only a small fraction of total centered-response variance, whereas model family and organization explain substantially more structure. (g) PC-wise branch AUC in raw V asks whether any individual raw-score PC acts as an AR-versus-MLM separation axis. (h) PC-wise branch AUC in V_d repeats this diagnostic after GLMap normalization. AUC values are direction-symmetrized as $\max(\text{AUC}, 1 - \text{AUC})$, so values near 0.5 indicate little branch separation and values near 1.0 indicate strong separation. Together, these diagnostics indicate that AR and MLM scores should not be interpreted as directly comparable raw probabilities, but their clipped and double-centered GLMap response profiles can be jointly visualized and clustered because branch identity is detectable but not the dominant source of variance in V_d .